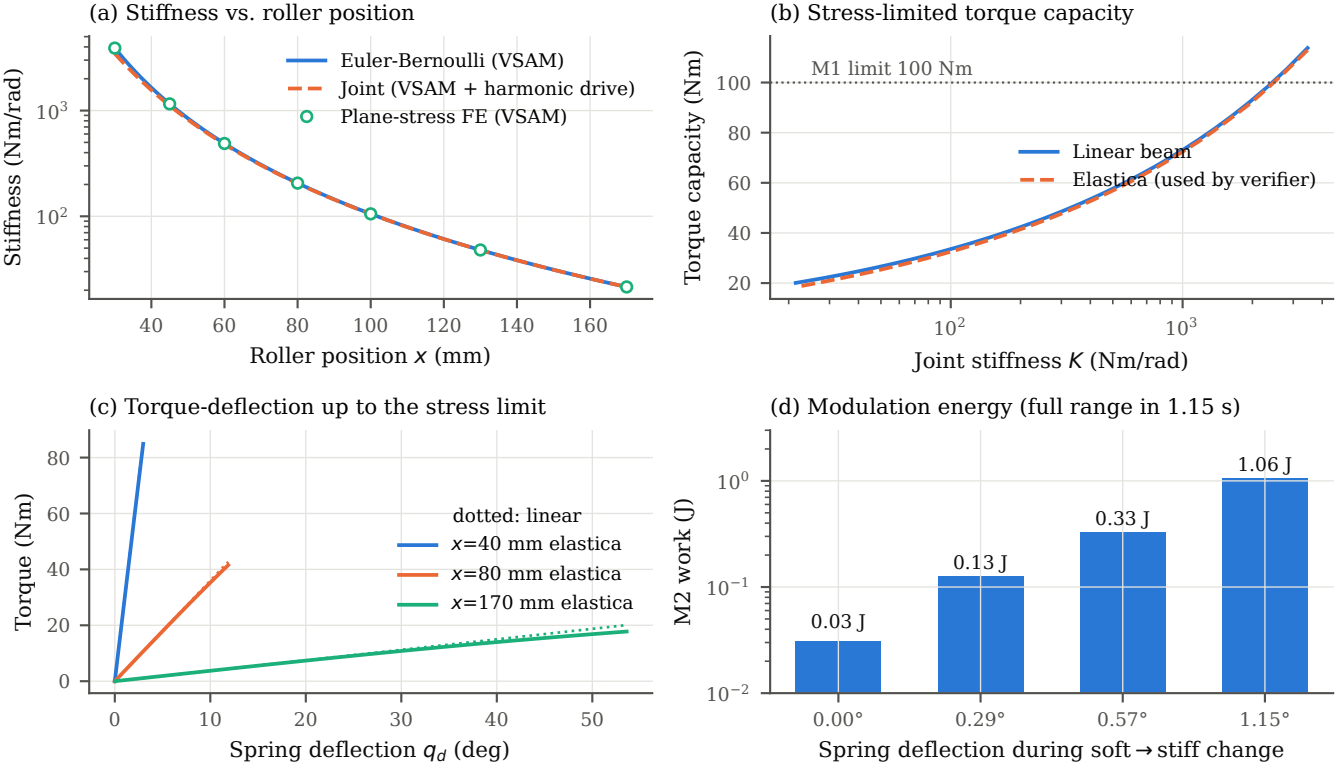
(a) Stiffness vs. roller position
Euler-Bernoulli (VSAM)
Joint (VSAM + harmonic drive)
Plane-stress FE (VSAM)
Stiffness (Nm/rad)
10^3
10^2
40
60
80
100
120
140
160
Roller position x (mm)
(b) Stress-limited torque capacity
M1 limit 100 Nm
Linear beam
Elastica (used by verifier)
Torque capacity (Nm)
100
80
60
40
20
10^2
10^3
Joint stiffness K (Nm/rad)
(c) Torque-deflection up to the stress limit
dotted: linear
x=40 mm elastica
x=80 mm elastica
x=170 mm elastica
Torque (Nm)
80
60
40
20
0
0
10
20
30
40
50
Spring deflection q_d (deg)
(d) Modulation energy (full range in 1.15 s)
M2 work (J)
10^0
10^{-1}
10^{-2}
0.03 J
0.13 J
0.33 J
1.06 J
0.00°
0.29°
0.57°
1.15°
Spring deflection during soft → stiff change